Jihong Min — List of Publications

Presidential Young Professor, Department of Biomedical Engineering
National University of Singapore
Google Scholar
28 papers (9 as first/co-first author), 8,718 total citations, h-index 25, updated April 2026.
† indicates equal contributions

Selected Publications

1. **Min, J.**†, Ahn, H.†, Lukas, H., Ma, X., Bhansali, R., Sunwoo, S.-H., Wang, C., Xu, Y., Yao, R., Kim, G., Li, Z., Hsiai, T. K., Emami, A., Jung, H.-T., & Gao, W. (2025). Continuous biochemical profiling of the gastrointestinal tract using a multiparametric ingestible capsule. ***Nature Electronics***, 8, 844-855. **Featured on Journal Cover** *(35 citations)*
2. Heng, W., Yin, S., **Min, J.**, Wang, C., Han, H., Shirzaei Sani, E., Li, J., Song, Y., Rossiter, H. B., & Gao, W. (2024). A smart mask for exhaled breath condensate harvesting and analysis. ***Science***, 385, 954-961. *(154 citations)*
3. **Min, J.**†, Demchyshyn, S.†, Sempionatto, J. R., Song, Y., Hailegnaw, B., Xu, C., Yang, Y., Solomon, S., Putz, C., Lehner, L., Schwarz, J. F., Schwarzinger, C., Scharber, M., Shirzaei Sani, E., Kaltenbrunner, M., & Gao, W. (2023). An autonomous wearable biosensor powered by a perovskite solar cell. ***Nature Electronics***, 6, 630-641. **Featured on Journal Cover** *(347 citations)*
4. **Min, J.**†, Tu, J.†, Xu, C.†, Lukas, H.†, Shin, S., Yang, Y., Solomon, S. A., Mukasa, D., & Gao, W. (2023). Skin-interfaced wearable sweat sensors for precision medicine. ***Chemical Reviews***, 123, 5049-5138. **Featured on Journal Cover** *(624 citations)*
5. Tu, J., **Min, J.**, Song, Y., Xu, C., Li, J., Moore, J., Hanson, J., Hu, E., Parimon, T., Wang, T.-Y., Davoodi, E., Chou, T.-F., Chen, P., Hsu, J. J., Rossiter, H. B., & Gao, W. (2023). A wireless patch for the monitoring of C-reactive protein in sweat. ***Nature Biomedical Engineering***, 7, 1293-1306. *(368 citations)*
6. **Min, J.**, Song, Y., & Gao, W. (2022). Microcracked conductors for wearable sensors. ***Nature Electronics***, 5(11), 717-718. *(9 citations)*
7. Wang, M.†, Yang, Y.†, **Min, J.**†, Song, Y., Tu, J., Mukasa, D., Ye, C., Xu, C., Heflin, N., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2022). A wearable electrochemical biosensor for the monitoring of metabolites and nutrients. ***Nature Biomedical Engineering***, 6, 1225-1235. **Featured on Journal Cover** *(861 citations)*
8. **Min, J.**†, Sempionatto, J. R.†, Teymourian, H.†, Wang, J., & Gao, W. (2021). Wearable electrochemical biosensors in North America. ***Biosensors and Bioelectronics***, 172, 112750. *(291 citations)*
9. Song, Y.†, **Min, J.**†, Yu, Y., Wang, H., Yang, Y., Zhang, H., & Gao, W. (2020). Wireless battery-free wearable sweat sensor powered by human motion. ***Science Advances***, 6(40), eaay9842. *(656 citations)*
10. **Min, J.**, Yang, Y., Wu, Z., & Gao, W. (2020). Robotics in the gut. ***Advanced Therapeutics***, 3(4), 1900125. *(104 citations)*

Other Publications

1. Heng, W., Putz, C., Tang, W., **Min, J.**, Lehner, L., Demchyshyn, S., Kim, G., Wang, C., Hailegnaw, B., Rippatha, E., Scharber, M., Kim, M., Chen, Y., Fan, Z., Kaltenbrunner, M., Gao, W. (2026). A battery-free smart mask for long-term exhaled breath biochemical sensing. ***Nature Sensors***, 1, 328-340. *(0 citations)*
2. Shin, S., Liu, R., Yang, Y., Lasalde-Ramírez, J. A., Kim, G., Won, C., **Min, J.**, Wang, C., Fan, K., Han, H., Uwakwe, C., Heng, W., Hsiai, T. K., Li, Z., FitzGerald, J. D., & Gao, W. (2025). A bioinspired microfluidic wearable sensor for multiday sweat sampling, transport, and metabolic analysis. ***Science Advances***, 11(33), eadw9024. *(37 citations)*
3. Wang, C., Fan, K., Shirzaei Sani, E., Lasalde-Ramírez, J. A., Heng, W., **Min, J.**, Solomon, S. A., Wang, M., Li, J., Han, H., Kim, G., Shin, S., Seder, A., Shih, C.-D., Armstrong, D. G., & Gao, W. (2025).

A microfluidic wearable device for wound exudate management and continuous in situ analysis of wound biomarkers. ***Science Translational Medicine***, 17(795), eadt0882. *(51 citations)*

4. Tu, J., Yeom, J., Chaj Ulloa, J., Solomon, S. A., **Min, J.**, Heng, W., Kim, G., Dao, J., Vemu, R., Pang, M., Wang, C., Kim, D.-H., & Gao, W. (2025). Stressomic: A wearable microfluidic biosensor for dynamic profiling of multiple stress hormones in sweat. ***Science Advances***, 11(32), eadx6491. *(35 citations)*
5. Wang, M., Ye, C., Yang, Y., Mukasa, D., Wang, C., Xu, C., **Min, J.**, Solomon, S. A., Tu, J., Shen, G., Tang, S., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2024). Printable molecule-selective core-shell nanoparticles for wearable and implantable sensing. ***Nature Materials***, 24, 589-598. *(72 citations)*
6. Xu, C., Song, Y., Sempionatto, J. R., Solomon, S. A., Yu, Y., Nyein, H. Y. Y., Tay, R. Y., Li, J., Heng, W., **Min, J.**, Lao, A., Hsiai, T. K., Sumner, J. A., & Gao, W. (2024). A physicochemical-sensing electronic skin for stress response monitoring. ***Nature Electronics***, 7, 168-179. *(345 citations)*
7. Ye, C., Wang, M., **Min, J.**, Tay, R. Y., Lukas, H., Sempionatto, J. R., Li, J., Xu, C., & Gao, W. (2024). A wearable aptamer nanobiosensor for non-invasive female hormone monitoring. ***Nature Nanotechnology***, 19, 330-337. *(402 citations)*
8. Song, Y., Tay, R. Y., Li, J., Xu, C., **Min, J.**, Shirzaei Sani, E., Kim, G., Heng, W., Kim, I., & Gao, W. (2023). 3D-printed epifluidic electronic skin for machine learning-powered multimodal health surveillance. ***Science Advances***, 9(37), eadi6492. *(228 citations)*
9. **Min, J.**, Gao, W. (2023). Battery-Free Wearable Electrochemical Sweat Sensors. ***IEEE IFETC***. *(1 citation)*
10. Choi, Y., Ho, D. H., Kim, S., Choi, Y. J., Roe, D. G., Kwak, I. C., **Min, J.**, Han, H., Gao, W., & Cho, J. H. (2023). Physically defined long-term and short-term synapses for the development of reconfigurable analog-type operators capable of performing health care tasks. ***Science Advances***, 9(27), eadg5946. *(53 citations)*
11. Shirzaei Sani, E., Xu, C., Wang, C., Song, Y., **Min, J.**, Tu, J., Solomon, S. A., Li, J., Banks, J. L., & Armstrong, D. G. (2023). A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds. ***Science Advances***, 9(12), eadf7388. *(365 citations)*
12. Mukasa, D., Wang, M., **Min, J.**, Yang, Y., Solomon, S. A., Han, H., Ye, C., & Gao, W. (2023). A Computationally Assisted Approach for Designing Wearable Biosensors toward Non-invasive Personalized Molecular Analysis. ***Advanced Materials***, 35(35), 2212161. *(89 citations)*
13. Yu, Y., Li, J., Solomon, S. A., **Min, J.**, Tu, J., Guo, W., Xu, C., Song, Y., & Gao, W. (2022). All-printed soft human-machine interface for robotic physicochemical sensing. ***Science Robotics***, 7(67), eabn0495. **Featured on Journal Cover** *(360 citations)*
14. Torrente-Rodríguez, R. M., Lukas, H., Tu, J., **Min, J.**, Yang, Y., Xu, C., Rossiter, H. B., & Gao, W. (2020). SARS-CoV-2 RapidPlex: a graphene-based multiplexed telemedicine platform for rapid and low-cost COVID-19 diagnosis and monitoring. ***Matter***, 3(6), 1981-1998. **Featured on Journal Cover** *(512 citations)*
15. Torrente-Rodríguez, R. M., Tu, J., Yang, Y., **Min, J.**, Wang, M., Song, Y., Yu, Y., Xu, C., Ye, C., & IsHak, W. W. (2020). Investigation of cortisol dynamics in human sweat using a graphene-based wireless mHealth system. ***Matter***, 2(4), 921-937. *(487 citations)*
16. Yang, Y., Song, Y., Bo, X., **Min, J.**, Pak, O. S., Zhu, L., Wang, M., Tu, J., Kogan, A., & Zhang, H. (2020). A laser-engraved wearable sensor for sensitive detection of uric acid and tyrosine in sweat. ***Nature Biotechnology***, 38(2), 217-224. *(1273 citations)*
17. Yu, Y., Nassar, J., Xu, C., **Min, J.**, Yang, Y., Dai, A., Doshi, R., Huang, A., Song, Y., & Gehlhar, R. (2020). Biofuel-powered soft electronic skin with multiplexed and wireless sensing for human-machine interfaces. ***Science Robotics***, 5(41), eaaz7946. *(694 citations)*
18. Song, Y., **Min, J.**, & Gao, W. (2019). Wearable and implantable electronics: moving toward precision therapy. ***ACS Nano***, 13(11), 12280-12286. *(252 citations)*